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Youth-like brain activation linked with greater cognitive training gains in older adults: Insights from the ACTOP study



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ABSTRACT

This study investigates the relationship between inter-individual neurofunctional differences in older adults and cognitive training efficacy, with a specific focus on the association between youth-like task-related brain activation and improvements in working memory (WM) training. The data is part of the Attentional Control Training for Older People (ACTOP) study, 30 older adults completed 12 half-hour WM training sessions. The WM performance slope, assessed at the conclusion of sessions 1 through 6 and sessions 7 to 12, determined early- and late-stage training gains, respectively. Transfer measures were taken before (PRE), midway (MID), and after (POST) training, and the differences in MID-PRE and POST-MID on transfer tasks were used to determine early- and late-stage transfer effects, respectively. The Goodness of Fit (GOF) metric was used to quantify the similarity between each older adult's activation pattern, as measured with functional magnetic resonance imaging (fMRI), to that of a group of younger adults. GOF scores were calculated for activation during low-load (1-0back) and high-load (2-0back) N-back tasks. The results indicated that larger GOF scores in the low-load condition were associated with greater training gains in both the early and late learning stages, and that larger GOF scores in the high-load condition were associated with greater training gains during the late-stage. These findings suggest that a youth-like brain activation pattern in older adults is associated with greater cognitive training benefits, underscoring the role of inter-individual neurofunctional differences to account for variations in training outcomes among older adults.

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1. Introduction

Aging is accompanied by changes in several memory domains (Desgranges, Eustache, & Rioux, 1994; Giffard, Desgranges, & Eustache, 2001). One of these domains is working memory. Many studies have reported lower working memory (WM) performance in older adults compared to their younger counterparts (Craig, Morris, & Gick, 1990; Reuter-Lorenz & Park, 2010; Sylvain-Roy, Lungu, & Belleville, 2015). Given the crucial role of WM in various complex cognitive activities, its decline in older individuals has been suggested to negatively impact both cognitive functioning and daily life (Vaughan & Giovanello, 2010). Many authors have proposed the use of cognitive training to enhance the WM of older adults, potentially leading to significant cognitive benefits. Most WM training programs adopt a process-based approach offering adaptive and repeated exercises aimed at improving the foundational processes underlying WM deficits (Boujut & Belleville, 2019; Matysiak, Kroemeke, & Brzezicka, 2019).

Several studies have indicated improvements resulting from WM training programs when assessing group performance in older adults (Boujut, Verty, et al., 2020; Dahlin et al., 2008; Li et al., 2008; Matysiak et al., 2019; for meta-analysis see Karbach & Verhaeghen, 2014). However, substantial differences exist at the individual level (Borella, Carbone, Pastore, De Beni, & Carretti, 2017; Matysiak et al., 2019). For example, Matysiak et al. (2019) reported that, after 25 sessions of adaptive WM training, 21.4% of older participants did not surpass their initial baseline performance level (Matysiak et al., 2019). Therefore, while WM training benefits are shown in many older adults, some individuals do not seem to exhibit improvement.

A few studies have attempted to understand whether these inter-individual differences in training efficacy are attributed to individual factors (for reviews see Teixeira-Santos et al., 2019; Ophey, Roheger, Folkerts, Skoetz, & Kalbe, 2020). Identifying a coherent set of underlying factors to explain inter-individual differences holds crucial clinical and theoretical implications (Desgranges, Baron, & Eustache, 1998). On a clinical level, it could help maximize efficacy for responders while offering alternative training approaches for non-responders, facilitating more personalized and efficient interventions. From a theoretical perspective, cognitive training can serve as an interesting experimental model for investigating brain plasticity processes in older adults (Belleville, Mellah, Boller, & Ouellet, 2023). Identifying the brain characteristics and factors behind individual differences in training responses may shed light on the underlying brain mechanisms that help maintain cognition in late adulthood.

Notably, one of the key advancements in cognitive neuroscience research related to aging is the demonstration of substantial inter-individual variability in functional activation changes with age (Cabeza, 2002; Nyberg & Bäckman, 2011). These observed differences may serve as explanatory factors for individual variations in cognitive training outcomes (for review see Baykara, Könen, Unger, & Karbach, 2021). A crucial inter-individual difference lies in the concept of maintenance, which refers to the brain's ability to retain its neural resources as individuals age (Nyberg & Bäckman, 2011). According to maintenance models, variations in age-related cognitive changes correspond to differences in brain alterations, with a more youth-like activation pattern associated with better cognitive performance (Fandakova, Lindenberger, & Shing, 2015; Nagel et al., 2009, 2011; Nyberg & Bäckman, 2011). The notion of compensation can also contribute to a better understanding of interindividual differences in brain activation. Compensation refers to neuroplastic processes that come into play to adapt to changes in the brain, ultimately supporting cognitive performance (Cabeza, 2002; Reuter-Lorenz & Cappell, 2008). Accordingly, differences in activation patterns observed in some older adults when compared to younger ones might indicate underlying compensation. It is important to note that the concepts of maintenance and compensation are not necessarily antithetical. The CRUNCH model, for example, suggests that brain maintenance and compensation for brain decline are two possible paths to cognitive success in older adults (Reuter-Lorenz & Cappell, 2008). Hence, a youthlike activation might reflect either activation maintenance or lack of compensation. Given the potential implications of maintenance and compensation in explaining cognitive aging, a crucial question arises: Do inter-individual differences in youth-like functional activation explain inter-individual differences in WM training gains?

Two studies have offered supportive evidence suggesting that older adults with a youth-like activation pattern may be the most responsive to WM training. Heinzel et al. (2014) examined this question by examining activation network efficiency and capacity in 20 older adults through WM-load-dependent activation. Neural efficiency was defined as the ability to maintain lower activation levels during a low WM-load (1-back) condition, while neural capacity referred to the ability to increase activation in response to higher WM-load (e.g., 3-back) conditions compared to the low WM-load condition. Given that existing literature indicates that younger adults typically exhibit superior efficiency and capacity compared to older adults, this activation pattern was interpreted as reflecting a more youth-like activation pattern. The authors observed that both greater efficiency

and greater capacity of the frontoparietal network were associated with higher behavioral gains in older adults after a 12-session WM training program (Heinzel et al., 2014). Vermeij et al. (2017) reported similar results in 30 older adults with and without mild cognitive impairment (MCI) using near-infrared spectroscopic imaging before a 25-session WM training program.

While these two studies suggest that maintaining a youth-like activation pattern in WM-related regions among older adults is associated to greater gains in WM training, several critical questions still require further investigation. When considering individual differences, a major challenge lies in identifying a single and meaningful metric to reflect the degree of youth-like activation at the individual level. For example, in the two previous studies, the authors did not compare a particular older participant to a group of younger adults to establish his/her youth-like pattern. Instead, they used scores reflecting how closely the activation difference between the high and low load WM conditions in a particular older participant resembled the pattern observed in younger adults (Heinzel et al., 2014; Nagel et al., 2011; Vermeij et al., 2017). Thus, the youth-like scores obtained by older adults expressed how similar their activation pattern was to the activation pattern observed or expected in younger adults, rather than being computed from an actual comparison with the scores of younger adults.

The Goodness of Fit method (GOF; Düzel, Schütze, Yonelinas, & Heinze, 2010), employed in this study, offers a promising approach to enhance the interpretation of inter-individual functional activation differences in the context of a youth-like activation pattern. This method assesses the degree of youth-like activation by comparing an individual older adult's activation pattern and the WM neurofunctional network observed in a reference group of younger adults who underwent the same fMRI protocol under similar conditions. The GOF method considers differences in activation both within the regions typically activated by younger adults and in regions outside of these areas. This is advantageous compared to previous approaches, as compensation models suggest that older adults may compensate by activating brain regions not engaged by younger adults (Cabeza et al., 2018). Notably, the two earlier studies selected regions of interest (ROIs) based on those reported in the literature for comparable experiments, thus excluding activation outside of pre-determined WM-related areas, limiting their interpretation of the observed youth-like pattern.

Another critical consideration is the timeline of training gains, which typically follows a non-linear pattern characterized by an initial stage of rapid improvement followed by a slower learning stage (Belleville et al., 2018; Belleville et al., 2022; Belleville, Mellah, et al., 2023; Boujut, Verty, et al., 2020; McAvinue et al., 2013). Learning models propose that these early and late learning stages rely on different cognitive and brain mechanisms (Chein & Schneider, 2012) and may be influenced by distinct individual characteristics (Belleville et al., 2022). The two previously mentioned studies focused on whether a youth-like activation was associated with change from pre-training to post-training, involving a relatively large number of training sessions. It is plausible that different mechanisms explain inter-individual differences in

WM training efficacy in the early rapid stage vs the later stage of training. Therefore, this study aimed to investigate which neurofunctional activation is associated with gains during either the early- or late-stage of training.

Lastly, an additional aim was to examine whether neurofunctional characteristics of older participants could be associated with inter-individual differences in transfer outcomes. This could offer valuable insights, as transfer to untrained tasks is often not observed following WM training (Dahlin et al., 2008; Ji, Wang, Chen, Du, & Zhan, 2016; Lawlor-Savage & Goghari, 2016). Similar to training gains, the extent of transfer gains might depend on inter-individual variability in baseline activation.

The primary objective of this study was to use the GOF metric, which quantifies youth-like activation at the individual level, to investigate whether youth-like functional activation in older adults is associated with greater WM training gains. Additionally, we aimed to determine whether a similar association is found for the early- and late-stage of training. The GOF value was calculated during the performance of the N-back task under low and high task loads. This was done to allow interpretation in terms of efficiency and capacity. Efficiency is typically examined under lower task load conditions. This is because lower levels of activation to complete the task under low load conditions is interpreted as reflecting better efficiency. Conversely, capacity is typically examined under high task load conditions, as higher recruitment under demanding conditions is typically interpreted as reflecting superior capacity (Barulli & Stern, 2013). Finally, a secondary objective was to examine whether youth-like activation is associated with greater transfer gains and, once again, whether this applies to the early- and late-stage of training.

Based on previous findings, we hypothesized that a larger GOF, reflecting a more youth-like brain activation, would be associated with greater training gains. While the specific effects pertaining to the early or late training stages are unknown, we expected a stronger association between brain activation and training gains in the early-stage given that training gains have been found to be more pronounced in the early-stage and dependant on individual characteristics. In younger adults, lower activation during lower WM task load reflects efficiency, whereas higher activation during high WM task load, reflects capacity. Consequently, a positive correlation between training gains and youth-like neurofunctional activation patterns at low WM task load in older adults was hypothesized as a link between neural efficiency and training gains. In contrast, a positive association between training gains and youth-like neurofunctional activation patterns at higher WM task loads was hypothesized as a link between greater neural capacity and training gains. Finally, we hypothesized that older adults with a youth-like activation pattern would exhibit greater transfer gains.

2. Methods

2.1. Transparency and openness

We report below how we determined our sample size, all data exclusions, all inclusion/exclusion criteria, whether

inclusion/exclusion criteria were established prior to data analysis, all manipulations, and all measures in the study. The analytic code and materials used for this research are available on Open Science Framework (OSF) at the following link; <https://osf.io/q29w8/>). Raw data cannot be publicly archived due to ethical considerations since the consent form explicitly states that the distribution and analysis of the data are limited to the principal investigator for additional projects. However, some but not all participants consented for their data to be used for further research. For inquiries about data sharing or more details, please reach out to the corresponding author. This research is embedded within the ACTOP randomized double-blind controlled trial, for which the study design was preregistered with the US National Institutes of Health clinical trial registry ([ClinicalTrials.gov](https://clinicaltrials.gov/ct2/show/NCT03532113) Identifier: NCT03532113); <https://clinicaltrials.gov/ct2/show/NCT03532113>. The learning timepoints, the fMRI analysis and the use of activation as a predictor of training efficacy were preregistered, but the GOF method was not.

2.2. Study design

The Attentional Control Training for Older People (ACTOP) study is a three-arm randomized controlled trial conducted at the Centre de recherche de l'Institut universitaire de gériatrie de Montréal (CRIUGM). Details on the study design and behavioral results can be found in [Boujut, Mellah, et al., 2020](#); [Boujut, Verty, et al., 2020](#). The study included healthy older adults who were randomly assigned to one of three training groups: WM updating, inhibition, and active control. For this study, we exclusively used data from the WM updating group since our focus was on individual differences in WM training efficacy. ACTOP participants received a functional magnetic resonance imaging (fMRI) examination one week prior to the initial training session (PRE), after the sixth session (MID), and after the twelfth session (POST). Given that our interest is on how baseline fMRI data predicts training gains and transfer effects, we solely considered neuroimaging data from the first scan session (PRE). To assess youth-like brain activation and the impact of age on cognition, a group of younger adults underwent an fMRI examination and behavioral assessment with no subsequent cognitive training. Training effect was measured using performance on each training session. Participants also underwent a behavioral assessment to measure transfer effects at PRE, MID and POST.

This was a double-blind study. The person responsible for randomization had no involvement in the enrollment process, cognitive assessments, or training. Research assistants conducting cognitive and fMRI evaluations were unaware of the participants' training conditions. Participants were informed that they would be randomized to different training programs, but they were unaware of the experimental and control conditions.

Ethics approval was granted by the Comité d'éthique de la recherche vieillissement-neuroimagerie of the Centre intégré universitaire de santé et de services sociaux du Centre-Sud-de-l'Île-de-Montréal, on May 8, 2017 and updated annually. Participants provided written consent after receiving a full explanation of the procedures.

2.3. Participants

The full ACTOP study's sample size was calculated with G*Power 3 software to ensure adequate power for detecting a small to medium effect size on the primary outcome (see [Boujut, Mellah, et al., 2020](#) for further details). In this study, participants consisted of 30 healthy older adults (aged 60 to 85) and 30 younger adults (aged 20 to 35) recruited in Montreal through local newspapers, associations, community centers, and the CRIUGM participant registry. All the included participants were right-handed, fluent French-speakers, and had normal or corrected vision and hearing. Older adults met inclusion criteria if they scored above the education-adjusted thresholds (≥ 9 for 16+ years of education; ≥ 5 for 8–15 years of education; ≥ 3 for 0–7 years of education) on the delayed recall subtest of the Wechsler Memory Scale Fourth Edition Logical Memory subtest ([Wechsler, 2009](#)). Participants were excluded if they exhibited indications of subjective cognitive decline (feeling that their memory was declining to the point of concern), had a history of or current central nervous system disease or injury, were taking psychoactive medication, contraindications and/or intolerability for MRI examination, neurological or psychiatric illness, or severe comorbidity. In addition to these criteria, younger adults were also excluded if they were full-time students. All inclusion and exclusion criteria were established prior to data analysis.

Demographic information (age, sex, and education level) was collected at baseline. For clinical assessments at baseline, participants completed questionnaires assessing vascular health (Hachinski Questionnaire; [Hachinski et al., 1975](#)), depressive symptoms (short version of the Geriatric Depression Scale for older adults and Beck Depression Inventory II for younger adults; [Yesavage, 1988](#); [Beck, Steer, & Brown, 1996](#)), general cognition (Montreal Cognitive Assessment; MoCA; [Nasreddine et al., 2005](#)), and episodic memory (Logical Memory subtest; [Wechsler, 2009](#)).

2.4. Working memory training

Training consisted of 12 30-min training sessions. Sessions were delivered three times per week for four weeks (one session per training day). The exercises were computerized and accessible on a Samsung Galaxy [Tablet 2](#) (Android version 4.2.2). Training took place at the research center in small groups of six to ten participants per group. The room was arranged with individual cubicles for participants to work on their own devices. A training supervisor was present to provide initial instructions and to address any technical issues that arose during the training.

The training included two versions of the N-back task, provided through the Neuropeak application ([Lussier, Saillant, Vrinceanu, Hudon, & Bherer, 2021](#)). Participants were presented with a series of items, each appearing for 4 s. For each item, they had to indicate whether it was the same or different from the item presented 1, 2, or 3 positions earlier (for the 1-back, 2-back, and 3-back condition, respectively). They conveyed their response by pressing the "=" (for same) or "≠" (for different) buttons on the right side of the screen using their thumb. The two versions of the N-back task included numbers (1–9) or symbols (snake, dog, planet, moon, star,

bird). Each session included both versions, presented in eight blocks with increasing WM load: 2 blocks of 1-back (11 items each), 3 blocks of 2-back (12 items each), and 3 blocks of 3-back (13 items each). In each block, 40% of the items matched the comparator. The program provided feedback on participants' performance at the conclusion of each training session. The program was adaptive: participants gained access to the 3-back only when 2-back performance was at 75% accuracy or higher. If performance accuracy on the 2-back block fell below 75%, three 1-back blocks were presented instead of the more difficult 3-back blocks.

2.5. Training gains and transfer measures

To assess training gains, we analyzed the performance on the N-back tasks completed during the training sessions. For each session, we computed a composite training score by combining 1) the Z-score of the total number of correct responses throughout the entire session, and 2) the Z-score of the average 2-back reaction time for that session. We included only the 2-back condition because it was the only one that was completed by all participants for all sessions and because it showed sufficient variability. Performance from the group of older adults in the first session was used as a reference to calculate the Z-scores. Training gains were then determined for each participant by calculating the slope value across training sessions 1 to 6 for the early-stage and across sessions 7 to 12 for the late-stage of training. This separation was based on the median session. Consequently, the division between early and late stages was not contingent upon observed learning, but on the number of learning sessions administered. Furthermore, we intentionally divided the training stages into two phases of six sessions to ensure that the late stage captured a segment of the training curve where performance changes were still occurring. Transfer effects were assessed through the completion of Keep Track and Running Span tasks administered on a computer using E-Prime. In the Keep Track task, each trial involved the random presentation of 12 words, each belonging to one of four semantic categories (fruits, clothing, music, colors). Participants were asked to orally recall the last word of each category. In the Running Span task, lists of consonants were presented in various and random lengths (span-1, span+1, span+3, span+5). Participants were asked to orally and sequentially recall the last span-1 letters without prior knowledge of the list's length, which was customized based on each participant's span, measured before the task. A composite transfer score was calculated for each session (PRE, MID, POST) by averaging the Z-scores derived from the proportion of correct responses on the Keep Track and the Running Span tasks. Performance from the group of older adults at PRE served as references for computing the Z-scores. Transfer gains were then calculated using the following formulas: MID minus PRE for the early-stage and POST minus MID for the late-stage of transfer.

2.6. fMRI methods

2.6.1. Scanning parameters

Neuroimaging data were acquired at the Unité de Neuroimagerie Fonctionnelle (UNF) of the Institut universitaire de

gériatrie de Montréal on a Siemens 3 T Prisma Fit Magnetic Resonance Imaging (MRI) scanner. For anatomical images, a T1-weighted 3D-MPRAGE sequence was obtained for co-registration (TR/TE 2300/2.98 ms, Fa 9°, FOV = 256 × 256, matrix 256 × 256, voxels 1 mm³, 192 slices). Functional images were captured using the echo-planar imaging sequence (TR/TE 785/30 ms, Fa 54°, FOV 192 × 192, matrix 64 × 64, voxels 3 mm³, 39 slices, axial section, and ascending order) using the simultaneous multiband technique with an acceleration factor of 6.

2.6.2. fMRI task

The fMRI WM task was a letter version of the N-back task different from the one used in the training task. It comprised fifteen 32-s blocks presented in a pseudorandom fashion. It included a control block (0-back) and two experimental blocks (1-back and 2-back). In the 0-back block, participants were instructed to determine whether the presented letter was an "X". In the 1-back block, they had to determine if the letter was the same as the previous one, and in the 2-back block, they were asked to identify whether the letter matched the second-to-last presented letter. Considering the adaptive nature of the training, we restricted the fMRI WM task to the 1-back and 2-back conditions in the scanner. By doing so, we anticipated that all participants would have undergone training on these conditions. Furthermore, based on our experience, performing tasks in the scanner environment is more stressful than performing them outside of the scanner, leading to lower performance levels, particularly in older adults. We thus opted not to include the 3-back condition to have a sufficient number of correct trials and to reduce the stress related to the overall experience in the scanner. Prior to the fMRI session, participants completed a short practice session on a computer in a separate room of the neuroimaging unit to ensure they understood the instructions, were familiar with the presentation format and knew how to respond. During the scan, instructions were projected on the screen for 4 s at the start of each block. Each block presented 16 letters, each displayed for 500 msec, followed by a 1500 msec presentation of a fixation cross. Between blocks, a 16-s set point was used as a baseline condition. Participants could respond "yes" with the middle finger or "no" with the index finger of the right hand using an MRI-compatible response box. For each task condition, a D-Prime index was calculated to assess participants' performance according to the statistical model of Macmillan and Creelman (2004). This involved subtracting the false alarm rate from the correct detection rate, followed by a Z-transformation (Macmillan & Creelman, 2004).

2.6.3. Image processing and analyses

Data were analyzed using MATLAB R2018a (<http://www.mathworks.com>) and Statistical Parametric Mapping software (SPM12; <http://www.fil.ion.ucl.ac.uk/spm>). Preprocessing included slice timing corrections, realignment of brain volumes for head motion corrections, image normalization based on the Montreal Neurological Institute (MNI) template, and spatial smoothing with a Gaussian kernel with dimensions of 8 × 8 × 8 mm full width at half maximum (FWHM) to enhance the signal-to-noise ratio. First-level fixed effects analyses, employing a generalized linear model (GLM), were carried out

following a block protocol. Regressors were generated, each condition of the WM fMRI task represented as a block using a canonical hemodynamic response function. To eliminate low-frequency noise, a high-pass filter (128 s) was applied to the time series response. Two contrasts were used for analyzing N-back fMRI activation: 1-0back and 2-0back.

Whole-brain one-sample *t*-tests were then conducted for each contrast (1-0back, 2-0back) with a significance threshold of $p < .05$ familywise error (FWE) for the fMRI data from the younger adults. These *t*-tests were used to create masks for the GOF calculation and to provide support for interpreting the GOF values in terms of efficiency or capacity. To assess age differences at each contrast (1-0back, 2-0back), two-sample whole-brain *t*-tests were performed to compare older and younger adults.

2.6.4. Goodness-of-fit calculation

The GOF method was used to assess the correspondence between the activation pattern of the group of younger adults and that of each older adult at baseline. The results of the *t*-tests on whole-brain activation in young adults were used to generate two masks. The inside mask represented the positive task activation of younger adults (at $p < .05$ family-wise error (FWE)). The outside mask encompassed all voxels located outside the inside mask. Both an inside and outside mask were created for each contrast. The GOF value for a particular older adult represented the average *T*-value of all voxels within the activation network of younger adults minus the average *T*-value of voxels outside the activation network of younger adults. A larger GOF value indicated smaller differences in activation patterns when compared to younger adults, suggesting a more youth-like activation pattern.

2.7. Statistical analyses

As part of the preliminary analyses, two-sample *t*-tests were conducted to examine group differences in demographic information, clinical characteristics, and cognitive performance

at baseline between older and younger adults. A paired one-sample *t*-test was used to assess the performance difference between the early- and late-stages of gains within the group of older adults who received the training, and Pearson correlation analyses (two-tailed) were used to evaluate their associations. A paired one-sample *t*-test was also used to assess the difference between the 1-0back GOF and the 2-0back GOF within the group of older adults who received the training, with Pearson correlations analyses (two-tailed) employed to measure their associations.

To determine if a youth-like pattern of activation was associated with greater training gains, Pearson correlations (two-tailed) were calculated between GOF values and training gains, separately for the early- and late-stages. Similarly, to investigate whether a youth-like pattern of activation was associated with greater transfer gains, Pearson correlations (two-tailed) were calculated between GOF values and transfer gains, also separately for the early- and late-stages.

3. Results

3.1. Preliminary analyses

A total of nine participants were excluded from the analyses. One older adult and four younger adults encountered problems during the fMRI examination, three older adults dropped out of the study before completion, and one younger participant was excluded due to an incidental finding. The final sample comprised 51 participants, consisting of 26 older adults (5 males) and 25 younger adults (6 males). Table 1 presents the means, standard deviations, and *t*-test results for the clinical characteristics and performance at baseline. As expected, older adults exhibited lower performance compared to younger adults on the MoCA, the Logical Memory subtest, the transfer measures, and the 2-back task completed during the fMRI. However, there were no age group differences in their level of education and performance on the 1-back task completed during the fMRI.

Table 1 – Participant characteristics.

Characteristics	Older Adults N = 26	Younger Adults N = 25	<i>p</i>
Demographic			
Sex, male/female	5/21	6/19	
Age (years)	68.0 (5.9)	27.4 (4.5)	<.001
Education (years)	14.5 (2.4)	15.4 (2.4)	.21
Clinical assessment			
MoCA (score, range 0–30)	28.2 (1.6)	29.1 (1.3)	.04
Delayed Logical Memory (score, range 0–25)	14.4 (3.8)	17.8 (3.5)	.002
Hachinski (score, range 0–18)	.3 (.6)		
GDS (score, range 0–15)	1.8 (2.9)		
BDI (score, range 0–63)		5.6 (2.5)	
Transfer measure at baseline			
WM composite (z-score)	0 (.8)	.8 (.7)	<.001
N-back performance at baseline			
1-back (<i>d'</i>)	3.2 (1.0)	3.5 (.6)	.11
2-back (<i>d'</i>)	2.2 (.7)	2.9 (.8)	<.001

Note. Except for sex, data are mean (standard deviation). MoCA = Montreal Cognitive Assessment. GDS = Geriatric Depression Scale. BDI = Beck Depression Inventory. WM = Working Memory. *d'* = D-Prime index.

Table 2 – Early- and late-stage training and transfer gains in older adults.

	Early-Stage	Late-Stage	t-value	p
Training gains ^a	.14 (.10)	.03 (.04)	6.29	<.001
Transfer gains ^b	.25 (.76)	.11 (.71)	.50	.31

Note. Data are mean (standard deviation).

^a Training gains = slope value across training sessions 1 to 6 (early-stage) and 6 to 12 (late-stage).

^b Transfer gains = performance at MID minus performance at PRE (early-stage) and performance at POST minus performance at MID (late-stage).

3.2. Activation at baseline

Fig. 1 shows the whole-brain analyses of the 1-0back and 2-0back contrasts separately, for older adults and younger adults, revealing activation of the frontoparietal network typically involved in WM processes (Owen, McMillan, Laird, & Bullmore, 2005). The regions activated by younger adults and older adults are detailed in Supplementary Tables S1 and S2. The whole-brain one-sample t-test analysis in younger adults revealed frontoparietal activations at 1-0back, $t(24) = 6.20$, $p < .001$, and 2-0back, $t(24) = 6.11$, $p < .001$. Similarly, the whole-brain one-sample t-test analysis in older adults showed frontoparietal activations at 1-0back, $t(25) = 6.01$, $p < .001$, and 2-0back, $t(25) = 5.87$, $p < .001$. In both groups, the 2-0back contrast exhibited higher t-values and larger clusters of activation compared to the 1-0back contrast. When comparing younger and older adults in a whole-brain two-sample t-test analysis, older adults displayed higher activation compared to younger adults for the 1-0back, $t(49) = 3.27$, $p < .001$, and for the 2-0back in frontoparietal regions, $t(49) = 3.27$, $p < .001$ (see appended Table S3).

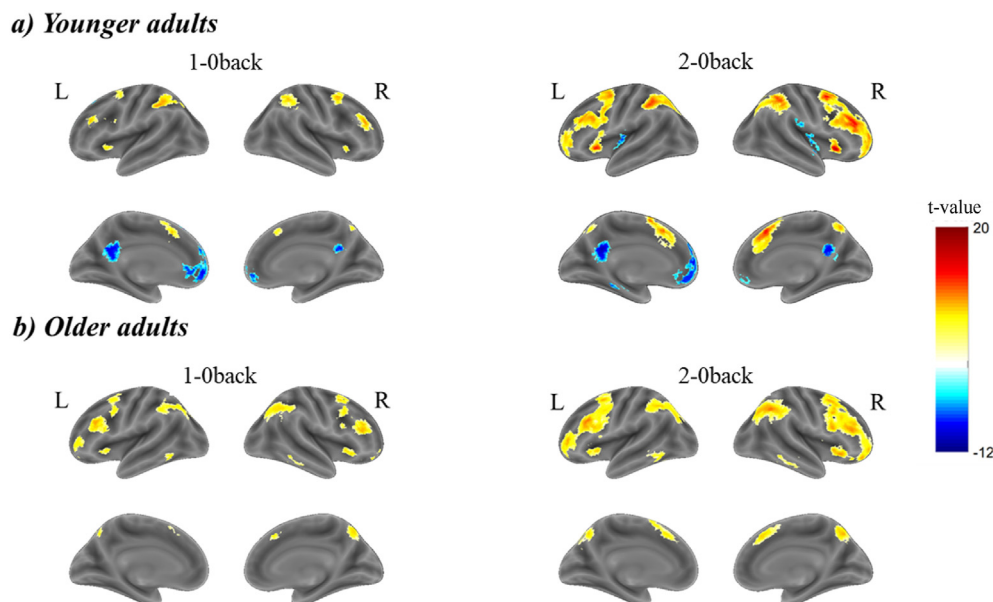
3.3. Training and transfer gain

Table 2 presents the training and transfer gains of older adults in both the early and late learning stages. Fig. 2 shows individual training score slopes for the training task. A paired one-sample t-test revealed that the slope of training gains was significantly higher in the early-stage (slope mean = .14; SD = .10) compared to the late-stage (slope mean = .03; SD = .04) of training, $t(25) = 6.29$, $p < .001$. The slopes of training gains for the early- and late-stages were positively correlated, $r(25) = .40$, $p = .04$. However, the paired one-sample t-test showed no significant difference between performance gains on the transfer task at the early-stage versus the late-stage, $t(25) = .50$, $p = .31$. Transfer gains in performance for the early- and late-stages were negatively correlated, $r(25) = -.75$, $p < .001$.

3.4. Goodness of fit effects and correlations with training and transfer gain

Table 3 shows the GOF values, condition differences, and correlations between 1-0back GOF and 2-0back GOF. The paired one-sample t-test revealed that the 2-0back GOF was significantly greater than the 1-0back GOF, $t(25) = -3.52$, $p = .002$. A positive correlation was observed between the 1-0back and the 2-0back GOF values, $r(25) = .72$, $p < .001$.

Table 4 illustrates the correlations between GOF values at both contrasts (1-0back, 2-0back) and training gains in the early and late-stages, as well as between the GOF values and transfer gains in both stages. Positive correlations were found between 1-0back GOF and training gains in both the early, $r(25) = .51$, $p < .01$, and late-stages of training, $r(25) = .44$, $p = .02$. Additionally, a significant positive correlation was noted between the 2-0back GOF and training gains at the late-stage, $r(25) = .42$, $p = .03$. These positive

**Fig. 1 – WM Activation at Baseline in Younger and Older Adults under the Two Load Conditions**

Note. Whole-brain one-sample t-tests with a significance threshold of $p < .05$ voxelwise FWE. T-score map activation (hot colors) and deactivation (cold colors). Cluster correction, $k > 20$.

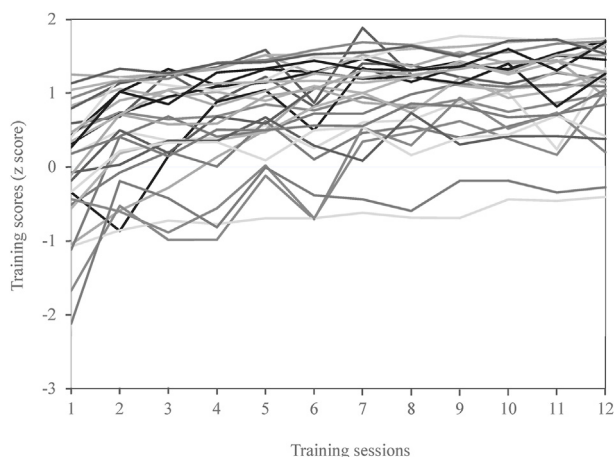


Fig. 2 – Individual Training Performance Scores for Older Participants Across Session 1 to 12

Note. The training score was computed by averaging the Z-scores of the number of correct responses for the entire session and the average reaction time Z-score for the 2-back task in that session.

correlations suggest that larger GOF values, indicating a more youth-like pattern, were associated with greater training gains. No other significant correlations were observed between GOF values and training gains. Notably, there were no significant correlations found between GOF values and transfer gains.

4. Discussion

Older adults exhibit variations in their functional activation, which can account for inter-individual differences observed in their WM training efficacy (Borella et al., 2017; Matysiak et al., 2019). This study aimed to investigate whether a youth-like functional activation pattern in older adults is associated with greater or lesser WM training gains. A secondary objective was to examine whether it was associated with greater transfer gains. This study introduces several innovations: It employs an individualized approach to assess how older adults respond to WM training, uses the GOF metric, a single measure that quantifies the degree of similarity between each older adult's brain activation pattern and that of younger adults during the same WM-related tasks, and examines effects across different task loads and training stages.

Table 3 – Goodness of Fit (GOF) values for the two load contrasts. comparison and correlation.

GOF		Load Comparison		Correlation	
1-0back	2-0back	t-value	p	r	p
2.4 (1.1)	3.0 (1.1)	−3.52	<.001	.72	<.001

Note. GOF data are mean (standard deviation).

4.1. Youth-like activation pattern associated with greater training gains

A first finding related to our primary objective is the positive association between the GOF value and training gains. This implies that individuals with a more youth-like activation pattern benefit the most from the training. While our finding aligns with prior research (Heinzel et al., 2014; Vermeij et al., 2017), our approach builds upon previous one because Heinzel et al. (2014) and Vermeij et al. (2017) focused on activation within predetermined WM-related areas. The GOF values used here is a metric that considers the difference between activation inside and outside of a younger adult's network. This is important since aging typically involves the recruitment of additional brain regions, which should be considered when assessing whether an individual exhibits a youth-like activation pattern. Our finding specifically indicates that the combination of greater activation inside and lesser activation outside of the youth-like activation pattern is beneficial for the behavioral plasticity of older adults.

In terms of practical implications, these findings suggest that individuals with a more youthlike activation pattern may respond better to interventions specifically targeting WM. Thus, even high performers have room for improvement. This is aligned with the magnification model (Lövdén, Brehmer, Li, & Lindenberger, 2012), which suggests that cognitive training provides greater benefit in older adults with superior performance, amplifying interindividual differences. It is important to emphasize that the observed results may be specific to the WM training used in this study. It is possible that individuals with a less youthlike activation pattern may benefit more from other training approaches or larger doses, as suggested by our previous work in prefrail older adults (Belleville et al., 2023). Thus, while our study underscores the potential of cognitive training for older adults, it also highlights the importance of tailoring interventions to individual characteristics to optimize outcomes.

4.2. Consideration of task load and training stage

Another significant finding in this study is the distinction between early and late training stages in interaction with WM load. Positive correlations were observed between 1-0back GOF (representing lower WM load) and training gains in both

Table 4 – Correlation between Goodness of Fit (GOF) values and training and transfer gain at the early-stage and late-stage of learning.

	Training Gain				Transfer Gain			
	Early-Stage		Late-Stage		Early-Stage		Late-Stage	
Contrasts	r	p	r	p	r	p	r	p
1-0back	.51	.00	.44	.02	.34	.09	−.30	.14
GOF								
2-0back	.36	.08	.42	.03	.13	.54	−.13	.53
GOF								

the early- and late-stages. We also identified a positive association between the 2-0back task GOF value (representing higher WM load) and training gains, but only during the later training stage. These findings can be interpreted using the concepts of neural efficiency and neural capacity. Neural efficiency refers to the ability to maintain lower levels of brain activation, typically observed in tasks with lower demand (Barulli & Stern, 2013). In contrast, neural capacity relates to the ability to increase activation as cognitive demands intensify (Barulli & Stern, 2013). In essence, efficiency denotes the minimal neural activity required for functioning, while capacity represents the maximum neural activity accessible when facing greater functional demands or neuronal damage (McKenzie et al., 2022). Consistent with previous research, younger adults in our study exhibited lower overall brain activation levels and smaller clusters during the 1-0back task compared to their activation levels during the 2-0back task (Heinzel et al., 2014; Mattay et al., 2006; Nagel et al., 2011). This pattern suggests that younger adults can rely on essential brain regions for effective task performance (Haier, Siegel, Tang, Abel, & Buchsbaum, 1992). The higher GOF values observed in older adults during the 1-0back task can be interpreted as indicative of a pattern akin to the efficiency observed in younger adults when the task load is low. Conversely, the larger GOF values found in older adults during the 2-0back task indicate that those older adults exhibit greater capacity, similar to what is observed in younger adults. Thus, efficiency predicts training gains in both early and late-stages, whereas capacity exclusively predicts late-stage training gains.

The findings regarding the distinct associations of capacity and efficiency with the early and late learning stages align with the triarchic learning model, which posits the involvement of different cognitive and neurophysiological mechanisms during the initial and later phases of learning (Chein & Schneider, 2012). Our results support the presence of differing mechanisms, indicating that better efficiency at baseline enables individuals to maintain gains throughout the learning stages, while enhanced capacity sustains the ability to fully capitalize on the increased learning opportunities presented in the later stage of training.

Interestingly, the divergence in functional mechanisms underlying training gains during the initial and later stages does not imply that these benefits are reserved for different subsets of older adults. In fact, our study uncovered a positive correlation between early and late training gains, suggesting that individuals who experience significant improvements in the initial stage are more likely to continue progressing during extensive training. The early training gains can serve as a foundational step upon which these individuals can build, leading to cumulative training gains over time. Moreover, our findings revealed a positive association between the 1-0back task GOF value and the 2-0back task GOF value, implying the coexistence of efficiency and capacity in individuals, as previously reported (Heinzel et al., 2014). Given the roles of both efficiency and capacity in cognitive training improvement, this association suggests that older adults who benefit most from training whether early or late are those who exhibit an overall youth-like activation pattern across various cognitive loads.

4.3. Lack of associations with transfer gains

One notable finding in our study is the absence of significant correlations between youth-like activation patterns and transfer outcomes, regardless of the training stage. This suggests that while a more youth-like activation pattern is associated with larger WM training gains, it may not necessarily lead to cognitive advantages that extend beyond the specific tasks targeted during training. Several factors may account for these results. One plausible explanation is the limited transfer observed in our study, which may not provide sufficient variability for a reliable association to be observed. The challenge of finding transfer following cognitive training has been widely acknowledged in the literature (Ji et al., 2016; Lawlor-Savage & Goghari, 2016), making it difficult to establish a direct link with individual neurofunctional characteristics. Alternatively, it is conceivable that inter-individual differences in transfer are associated with different underlying processes than those related to training gains. For example, a previous study in younger and older adults using WM training found common striatum activation in the training and transfer tasks (Dahlin et al., 2008). This study suggested that successful transfer occurs when the transfer task engages overlapping processing components and brain regions with the trained tasks. Interestingly, the authors noted limited transfer in older adults, which they attributed to lower striatal activation before training. Thus, future studies could explore whether a GOF approach can be used to assess whether a youth-like activation pattern in brain regions activated by both the training and transfer tasks can predict successful transfer.

It is interesting to note that we found a negative correlation between transfer gains for the early- and late-stages. This negative correlation suggests a distinct pattern among participants: those who demonstrated the least transfer early on exhibited larger transfer later on, and conversely, those who improved most on transfer tasks early on showed lower level of late transfer. A likely explanation is that it reflects the fact that there would be fast learners and slow learners for transfer. The identification of different learning types based on pace is a common observation noted in the literature on learning and memory (Varghese & Aneesa, 2021). Another possibility is that this reflects regression to the mean, a common phenomenon in repeated design (Barnett, Van Der Pols, & Dobson, 2005).

4.4. Youth-like versus youthful pattern of activation

It is important to clarify the distinction between a youth-like activation pattern and what might be considered a youthful pattern. We use the term “youth-like activation” to describe the similarity between the activation patterns observed in older adults and those observed in younger individuals. Some might argue that such a pattern alone does not necessarily qualify as youthful, as it may not consider potential benefits of compensatory activation in older adults. Compensation models propose that older adults can maintain cognitive performance during high cognitive demand tasks by recruiting additional neural resources compared to younger adults (Cabeza, 2002; Reuter-Lorenz & Cappell, 2008). If the youth-like

activation pattern observed in this study indeed reflects a lack of compensation, our findings suggest that older adults who do not engage in compensatory mechanisms derive the most benefit from WM training.

Therefore, while a youth-like activation pattern may indicate the maintenance of youthful activation, it might also reflect the lack of compensation. However, there are several reasons supporting the view that the youth-like activation patterns observed here indicate greater maintenance (Nyberg & Bäckman, 2011) rather than a lack of compensation. One argument is rooted in numerous studies that have demonstrated that youth-like neurofunctional characteristics in older adults are associated with superior cognitive performance (Fandakova et al., 2015; Nagel et al., 2009, 2011; Nyberg & Bäckman, 2011). In a related analyses conducted by our team, using a slightly different sample but the same task, we similarly observed that a more youth-like activation pattern, as assessed by GOF, was associated with younger age, higher WM performance, and overall cognitive proficiency. Consequently, our findings suggest that individuals who maintain the integrity of their WM activation as they age stand to derive the most substantial advantages from WM training.

4.4.1. Strengths and limitations

This study is the first to examine the relationship between youth-like brain activity and training gains, using a GOF approach, a singular metric assessing the degree of correspondence between a younger adult's network and each older adult's activation pattern. We also used a data-driven approach to establish the task activation network of younger adults, ensuring that both age groups followed the same fMRI protocol, which reduces potential methodological biases. Furthermore, we are pioneers in investigating the association between baseline brain activation and early and late training stages, providing valuable insights into inter-individual neurofunctional variability in training response and contributing to the enrichment of existing knowledge in this domain.

One limitation of our study is that we assessed youth-like activation patterns in older adults at a single time point. While our approach does not allow for the continuous monitoring of the persistence of these differences in youth-like activation patterns during training, and although not our primary objective, it would be interesting to investigate whether training can induce or enhance youth-like activation. Another limitation is the relatively small sample size, which might have reduced our power to detect small effects, particularly for correlations. Additionally, our examination focused solely on functional activation, while other variables, such as brain structure and connectivity, may also be linked to cognitive training gains. An additional consideration arises from previous research which suggests that lifestyle and differences in health factors are associated with brain and cognitive differences (Desgranges et al., 1994; Arenaza-Urquijo et al., 2013; Chételat et al., 2017). Future research might consider the role education on the association between neurofunctional variability and cognitive training gains in older adults (e.g., Belleville, Mellah, de Boysson, Demonet, & Bier, 2014). Finally, we observed no associations with transfer effects, possibly due to the use of a global GOF measure or

the limited effect, and consequently, the limited variability in transfer.

5. Conclusion

In summary, our study has shown that older adults achieve better results in WM training when their baseline brain activation closely resembles that of younger adults. Specifically, a more youth-like activation pattern during the 1-0back task was associated with higher cognitive training gains in both the early and late-stages, while a similar pattern during the 2-0back task was associated with higher cognitive training gains in the late-stage. This study deepens our understanding of the relationship between youth-like neural efficacy and neural capacity, both of which contribute to inter-individual differences in WM training efficacy. As these findings result from WM training, future studies should investigate whether similar relationships extend to other cognitive training domains, such as episodic memory training.

Open practices

The study in this article has earned Open Materials and Pre-registered badges for transparent practices. The materials and preregistered studies are available at: <https://osf.io/q29w8/>.

CRediT authorship contribution statement

Lynn Valeyry Verty: Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **Samira Mellah:** Data curation, Formal analysis, Investigation, Project administration, Validation, Visualization, Writing – review & editing. **Samantha Maltezos:** Conceptualization, Data curation, Investigation, Writing – review & editing. **Arnaud Boujut:** Data curation, Investigation, Project administration, Writing – review & editing. **Maxime Lussier:** Resources, Software, Writing – review & editing. **Louis Bherer:** Resources, Writing – review & editing. **Sylvie Belleville:** Resources, Project administration, Methodology, Funding acquisition, Conceptualization, Supervision, Validation, Writing – original draft.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT and DeepL Translate to improve the readability and syntax. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

SB is a consultant for Lucilab for the validation a cognitive decline prevention program. She has intellectual property

rights on the following programs: “Programme de Stimulation pour une santé cognitive, Memoria, Batterie d’évaluation de la mémoire Côte-des-Neiges”, MEMO, Méthode d’Entraînement pour une Mémoire Optimale”, “Atelier de Stimulation Cognitive” “e-space” and “Brain Health PRO”. These programs are not used for commercial purposes. She collaborates with Mind Maze on a program combining physical activity and cognitive training. LB and ML have developed and have intellectual property rights for Neuropeak, a digital platform to evaluate and train performance on executive functioning tasks. This platform has not been used for commercial purposes.

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Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cortex.2024.04.013>.

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